IBM's India hardware journey

In 2024, a new class of AI infrastructure is powering the action. And increasingly, it's being engineered in India. With the recent launch of IBM's Power11 platform, the country isn't just consuming next-gen tech—it's building it for the world. "Power11 is engineered for the GenAI era where speed, uptime, and real-time inferencing aren't nice-to-haves; they're non-negotiable," says Subhathra Srinivasaraghavan, Vice President, IBM ISDL. Edited excerpts of our interview follow:

SUBHATHRA SRINIVASARAGHAVAN
Vice President, IBM ISDL

Q How IBM India Systems Development Labs help build Power11?

A Power11 is a global innovation, and India is at the core of it. Engineers at IBM India Systems Development Lab (ISDL) collaborated closely with worldwide teams to co-develop the platform across hardware, firmware, and software.
Indian talent played a key role in shaping Power11's AI-first design. With ISDL holding global mission ownership for different areas of Power, this launch reflects how India isn't just adopting next-gen infrastructure; it's actively building it for the world.

Q Challenges and hopes of India's silicon future?

A India's moment in silicon is now. With nearly 20% of the world's chip design talent, a thriving startup ecosystem, and bold national missions like India Semiconductor Mission and IndiaAI, the foundation is already in place. The challenge lies in scaling deep-tech capabilities, from design to fabrication, while building long-term industry-academia-government collaboration.
At ISDL, we have built great talent who own and design our future processors and are developing world-class EDA tools that power chip design globally. Our partnerships with IITs, CDAC, and L&T Semiconductor Technologies show that when India builds together, the world takes notice.

Q Your advice for youngsters to chart a career in deep tech in India?

A Skills are the new currency. In a world powered by AI, semiconductors, and quantum, there are no "non-tech" roles anymore. IBM is committed to skilling 30 million by 2030. If you're curious, start now. Learn by doing, explore and solve real problems.

projects which helped build credibility and attract top talent," Deepak says. "Today, the India team is not just a support arm, it's central to the AMD global design engine."

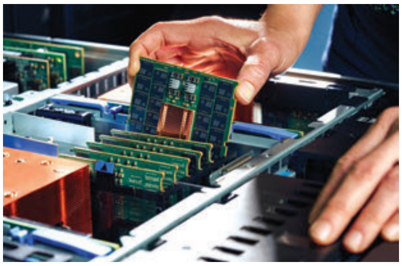
This evolution isn't just about hiring – it's about rethinking what India can be. And in many ways, Deepak has been a quiet catalyst for that change.

Of course, the road to building a world-class semiconductor design team in India hasn't been frictionless – especially in a domain as specialized as silicon engineering. The global shortage of semiconductor talent is very real, and India (while rich in raw numbers) still faces an acute skills gap in chip design, verification, and physical layout.

"Recruiting people who were not just technically sound but also aligned with AMD's values of innovation and collaboration was key," Deepak explains. "Structured onboarding, and continuous learning initiatives helped maintain the quality as we scaled."

It helps that the wind, at long last, seems to be blowing in India's direction. The govt's semiconductor mission and design-linked incentive schemes have attracted serious interest globally.

That said, there are caveats. While India has done well on the design side, the design-to-fab pipeline still faces friction. Local prototyping infrastructure remains limited, and most chip designs are still taped-out abroad, introducing delays and dependency.



"To bridge this, we need deeper collaboration between design houses, academia, and emerging foundries," Deepak cautions. "Investment in prototyping infrastructure and access to industry-grade EDA tools for academia can also play a critical role."

In many ways, AMD's story in India – and Deepak's career arc – mirrors the country's own ambition. Rooted in a belief that India can go from consumer to contributor in the global tech stack. From semiconductors and AI to the still-speculative realm of quantum computing, the narrative is shifting from 'Make in India' to 'Innovate in India.'

And there's no shortage of young minds eager to jump in. "My advice is to build a solid foundation in both hardware fundamentals and system-level thinking," Deepak offers to aspiring engineers. "Don't just chase jobs, chase problems worth solving." d